Cancer Survival ML Project | Pipeline Overview

- End-to-end pipeline: public TCGA-BRCA data → cloud storage → SQL parsing → statistical gene selection → neural network → dashboard.
- 1,208 patients narrowed to 389 with complete survival, stage and age records (325 tumor + 64 control samples).
- Each step is a numbered notebook in the GitHub repository

Key data decisions:

- Metastatic samples **excluded in the API query**.
- **11 patients** with several tumor samples **reduced to one**, so no patient appears in **both train and test**.
- GDC moved follow-up data to a new field, so I queried the new one.
- Patients with **unknown 5-year outcome** were **excluded**, not assumed to survive.
- Scaling and PCA were **fit on training data only**.

DATA MIGRATION

GDC API Call

Pull case + file metadata; exclude Metastatic samples at source

Python + requests

Call for metadata and RNA-seq data **(1,206 raw files gathered)**

S3 Upload

Stream raw RNA-seq count files to AWS S3

boto3 + AWS S3

Upload of raw RNA-seq files, **both Tumor and Control cohorts**

Metadata Parsing

DuckDB: unnest JSON, compute survival / stage / age

DuckDB SQL

389 profiles selected based on metadata's completeness

Parquet Conversion

Filter to protein-coding genes per sample

DuckDB + Parquet

Count files from s3 selected for protein-coding genes, and merged samples into one Parquet file/group.
Metric: 389 samples × 19,962 genes

DATA ANALYSIS & MODELING

DESeq2

Differential expression, tumor vs. control

PyDESeq2

Selection of 1,738 significant genes.
(adjusted $p < 0.05$, $|\log2FC| > 2$)

Feature Engineering

PCA (50 components) vs. DESeq2-selected genes

scikit-learn PCA

The expression input:
7 clinical metadata features with either **50 PCA components or DESeq2-selected genes.**

Neural Network

Dual-branch model, 5-fold cross-validation

PyTorch

The two branches (metadata | RNA-seq) **concatenated (96 units),** passed through **batch norm, ReLU** and **dropout,** then a **single output.**

Baseline Models

Random Forest & Logistic Regression comparison

scikit-learn

Logistic Regression | Random Forest: Randomized search **over 20 draws** with **5 fold CV,** scored on **accuracy**

Power BI Dashboard

Cohort profile + model performance

Power BI

Final deliverable